



Security & Data Protection

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Sensitive Data Discovery, Masking, and Compliance

This notebook covers sensitive data discovery, OpenPipeline masking configuration, security event monitoring, and compliance reporting.

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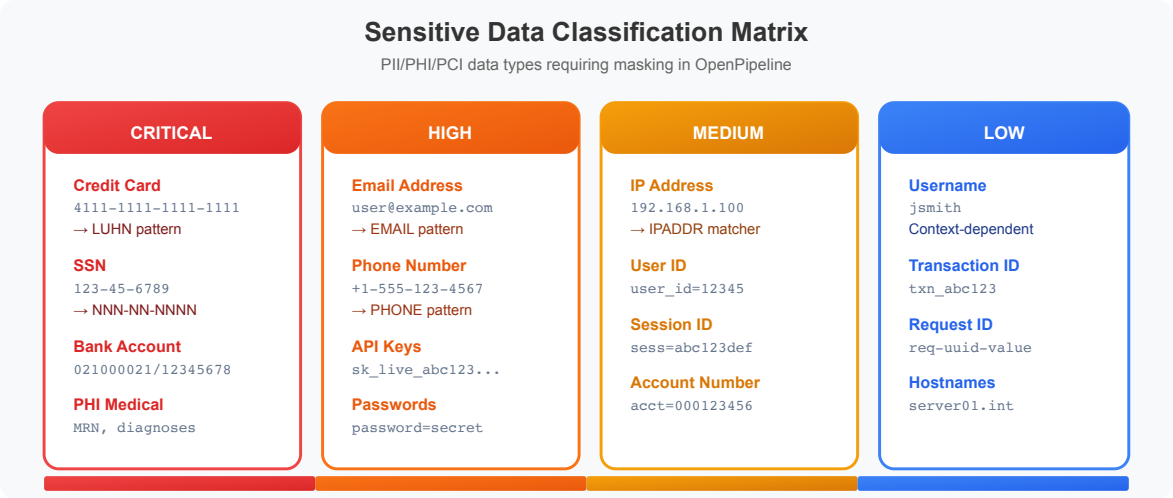
Prerequisites

-  Access to a Dynatrace environment with log data
-  Completed OPLOGS-01 through OPLOGS-07

-  Understanding of data privacy requirements

1. Sensitive Data Discovery

Before implementing masking, discover what sensitive data exists in your logs.



```
// Search for potential email addresses in logs
fetch logs, from: now() - 1h
| filter contains(content, "@")
| filter NOT contains(content, "@dynatrace")
| filter NOT contains(content, "@example")
| fieldsAdd content_preview = substring(content, from: 0, to: 150)
| summarize {count = count()}, by: {content_preview, k8s.namespace.name}
| sort count desc
| limit 20
```

```
// Search for potential API keys or tokens
fetch logs, from: now() - 1h
| filter contains(content, "key=")
      OR contains(content, "token=")
      OR contains(content, "api_key")
      OR contains(content, "apikey")
      OR contains(content, "secret")
| fieldsAdd content_preview = substring(content, from: 0, to: 120)
| summarize {count = count()}, by: {content_preview, k8s.namespace.name}
| sort count desc
| limit 20
```

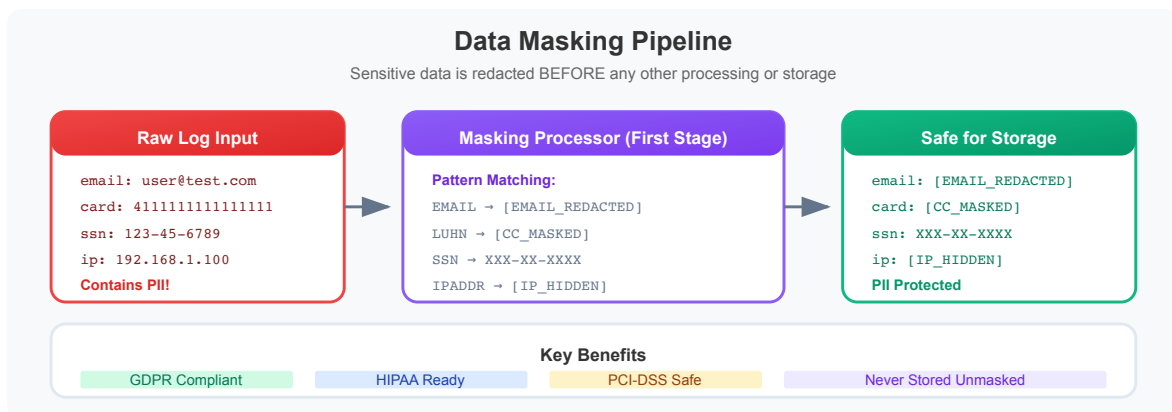
```
// Search for potential JWT tokens
fetch logs, from: now() - 1h
| filter contains(content, "eyJ") // JWT typically starts with eyJ
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview, k8s.namespace.name}
| sort count desc
| limit 15
```

```
// Search for potential credit card patterns (16 digits)
fetch logs, from: now() - 1h
| filter contains(content, "card")
      OR contains(content, "payment")
      OR contains(content, "visa")
      OR contains(content, "mastercard")
| fieldsAdd content_preview = substring(content, from: 0, to: 120)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 15
```

```
// Search for password-related entries (should never be logged!)
fetch logs, from: now() - 1h
| filter contains(content, "password")
      OR contains(content, "passwd")
      OR contains(content, "pwd=")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview, k8s.namespace.name}
| sort count desc
| limit 20
```

2. OpenPipeline Masking Configuration

OpenPipeline provides built-in masking processors to protect sensitive data at ingestion time.



Masking Processor Types

Processor	Use Case
Mask value	Replace with fixed pattern (e.g., <code>***MASKED***</code>)
Hash value	One-way hash for correlation without exposure
Remove field	Completely remove the field
Pattern mask	Partial masking (e.g., <code>****1234</code>)

Configuration Path

- Settings → Log processing → OpenPipeline

2. Select or create pipeline
3. Add "Mask value" or "Hash value" processor
4. Configure field and pattern

DPL Patterns for Masking

```
# Email pattern
[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}
```

```
# Credit card pattern
\b(?:\d[ -]*?){13,16}\b
```

```
# SSN pattern (US)
\b\d{3}-\d{2}-\d{4}\b
```

```
# IP address pattern
\b(?:\d{1,3}\.){3}\d{1,3}\b
```

```
// Verify masking is working (look for masked patterns)
fetch logs, from: now() - 1h
| filter contains(content, "MASKED")
      OR contains(content, "***")
      OR contains(content, "[REDACTED]")
| fieldsAdd content_preview = substring(content, from: 0, to: 120)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 15
```

```
// Check which pipelines are processing logs
fetch logs, from: now() - 1h
| summarize {count = count()}, by: {dt.openpipeline.pipelines}
| sort count desc
```

3. IP Address Analysis

IP addresses may require masking depending on your compliance requirements.

```
// Find logs containing IP addresses
fetch logs, from: now() - 1h
| parse content, "IPADDR:ip_found"
| filter isNotNull(ip_found)
| summarize {count = count()}, by: {ip_found, k8s.namespace.name}
| sort count desc
| limit 20
```

```
// Classify IP addresses (internal vs external)
fetch logs, from: now() - 1h
| parse content, "IPADDR:ip_found"
| filter isNotNull(ip_found)
| fieldsAdd ip_str = toString(ip_found)
| fieldsAdd ip_type = if(startsWith(ip_str, "10."), "RFC1918-10",
                        else: if(startsWith(ip_str, "192.168."), "RFC1918-192",
                        else: if(startsWith(ip_str, "172."), "RFC1918-172",
                        else: if(startsWith(ip_str, "127."), "LOOPBACK",
                        else: "EXTERNAL"))))
| summarize {count = count()}, by: {ip_type}
| sort count desc
```

4. Security Event Monitoring

Monitor logs for security-relevant events.

```
// Authentication-related logs
fetch logs, from: now() - 1h
| filter contains(content, "login")
      OR contains(content, "auth")
      OR contains(content, "signin")
      OR contains(content, "logout")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview, k8s.namespace.name}
| sort count desc
| limit 20
```

```
// Failed authentication attempts
fetch logs, from: now() - 1h
| filter (contains(content, "failed") OR contains(content, "denied") OR
contains(content, "unauthorized"))
      AND (contains(content, "login") OR contains(content, "auth") OR
contains(content, "access"))
| fieldsAdd content_preview = substring(content, from: 0, to: 120)
| summarize {count = count()}, by: {content_preview, k8s.namespace.name}
| sort count desc
| limit 20
```

```
// Security-related errors over time
fetch logs, from: now() - 24h
| filter loglevel == "ERROR"
| filter contains(content, "security")
      OR contains(content, "unauthorized")
      OR contains(content, "forbidden")
      OR contains(content, "denied")
| makeTimeseries {security_errors = count()}, interval: 30m
```

```
// Access pattern anomalies - hourly access distribution
fetch logs, from: now() - 24h
| filter contains(content, "access") OR contains(content, "request")
| fieldsAdd hour_bucket = bin(timestamp, 1h)
| summarize {access_count = count()}, by: {hour_bucket, k8s.namespace.name}
| sort hour_bucket asc
```

5. Audit Log Queries

Track administrative and configuration changes.

```
// Administrative action logs
fetch logs, from: now() - 24h
| filter contains(content, "created")
      OR contains(content, "deleted")
      OR contains(content, "updated")
      OR contains(content, "modified")
| filter contains(content, "user")
      OR contains(content, "admin")
      OR contains(content, "config")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {
    count = count(),
    first_seen = min(timestamp),
    last_seen = max(timestamp)
}, by: {content_preview, k8s.namespace.name}
| sort count desc
| limit 20
```

```
// Configuration change events
fetch logs, from: now() - 24h
| filter contains(content, "config")
      OR contains(content, "setting")
      OR contains(content, "environment")
| filter contains(content, "change")
      OR contains(content, "update")
      OR contains(content, "reload")
| fieldsAdd content_preview = substring(content, from: 0, to: 120)
| summarize {count = count()}, by: {content_preview, k8s.namespace.name}
| sort count desc
| limit 15
```

```
// Permission-related events
fetch logs, from: now() - 24h
| filter contains(content, "permission")
      OR contains(content, "role")
      OR contains(content, "privilege")
      OR contains(content, "rbac")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview, k8s.namespace.name}
| sort count desc
| limit 15
```


6. Compliance Reporting

```
// Data access summary
fetch logs, from: now() - 24h
| filter contains(content, "access") OR contains(content, "read") OR
contains(content, "view")
| summarize {
    total_accesses = count(),
    unique_sources = countDistinct(dt.openpipeline.source)
}, by: {k8s.namespace.name}
| sort total_accesses desc
| limit 15
```

```
// Log retention verification
fetch logs, from: now() - 7d
| summarize {
    earliest = min(timestamp),
    latest = max(timestamp),
    total_logs = count()
}, by: {dt.system.bucket}
| fieldsAdd retention_days = (latest - earliest) / 864000000000000
```

```
// Sensitive field exposure audit
fetch logs, from: now() - 1h
| summarize {
    total = count(),
    with_email_pattern = countIf(contains(content, "@")),
    with_key_pattern = countIf(contains(content, "key=") OR contains(content,
"token=")),
    with_password_ref = countIf(contains(content, "password") OR
contains(content, "passwd"))
}
| fieldsAdd email_exposure_pct = round((with_email_pattern * 100.0) / total,
decimals: 2)
| fieldsAdd key_exposure_pct = round((with_key_pattern * 100.0) / total,
decimals: 2)
| fieldsAdd password_exposure_pct = round((with_password_ref * 100.0) /
total, decimals: 2)
```

7. Masking Best Practices

Do's

1. **Mask at ingestion** - Use OpenPipeline processors to mask before storage
2. **Use hashing** - For fields that need correlation but not visibility
3. **Audit regularly** - Run discovery queries to find new sensitive data patterns
4. **Document patterns** - Keep a registry of masked data types and patterns
5. **Test in staging** - Verify masking works before production deployment

Don'ts

1. **Don't log passwords** - Ever, even "for debugging"
2. **Don't log full credit cards** - Maximum last 4 digits
3. **Don't rely on application masking alone** - OpenPipeline is your safety net
4. **Don't skip internal data** - Internal IP/usernames may still be sensitive
5. **Don't forget API keys** - Rotate if accidentally logged

OpenPipeline Masking Configuration Example

```
# OpenPipeline processor configuration
processors:
  - name: mask-emails
    type: mask
    field: content
    pattern: "[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.\.[A-Za-z]{2,}"
    replacement: "[EMAIL-MASKED]"

  - name: mask-api-keys
    type: mask
    field: content
    pattern: "(api_key|apikey|key)=[A-Za-z0-9]{20,}"
    replacement: "$1=[KEY-MASKED]"
```

```
// Security health summary
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    security_events = countIf(contains(content, "security") OR
contains(content, "auth")),
    failed_events = countIf(contains(content, "failed") OR contains(content,
"denied")),
    potential_pii = countIf(contains(content, "@") OR contains(content,
"password"))
}
| fieldsAdd security_event_rate = round((security_events * 100.0) /
total_logs, decimals: 2)
| fieldsAdd failure_rate = round((failed_events * 100.0) / total_logs,
decimals: 2)
| fieldsAdd pii_risk_rate = round((potential_pii * 100.0) / total_logs,
decimals: 2)
```



Summary

In this notebook, you learned:

- ✓ **Sensitive data discovery** - Finding PII, tokens, and credentials
 - ✓ **OpenPipeline masking** - Configuration and patterns
 - ✓ **IP address analysis** - Classification and masking needs
 - ✓ **Security monitoring** - Auth events, failures, anomalies
 - ✓ **Audit logging** - Admin actions, config changes
 - ✓ **Compliance reporting** - Data access and exposure audits
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Next Steps

Continue to **OPLOGS-07: Buckets & Cost Optimization** for storage management.

References

- [OpenPipeline Data Masking](#)
 - [Log Security Best Practices](#)
 - [GDPR and Compliance](#)
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